

Linux Class 6 - Practice Task

Practising users, groups and file permissions

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Date: 28 September 2026

For practice I made two new users, deepa and manoj, and one group called book-club. At the end I deleted all of them again, because deleting users and groups is also in the notes.

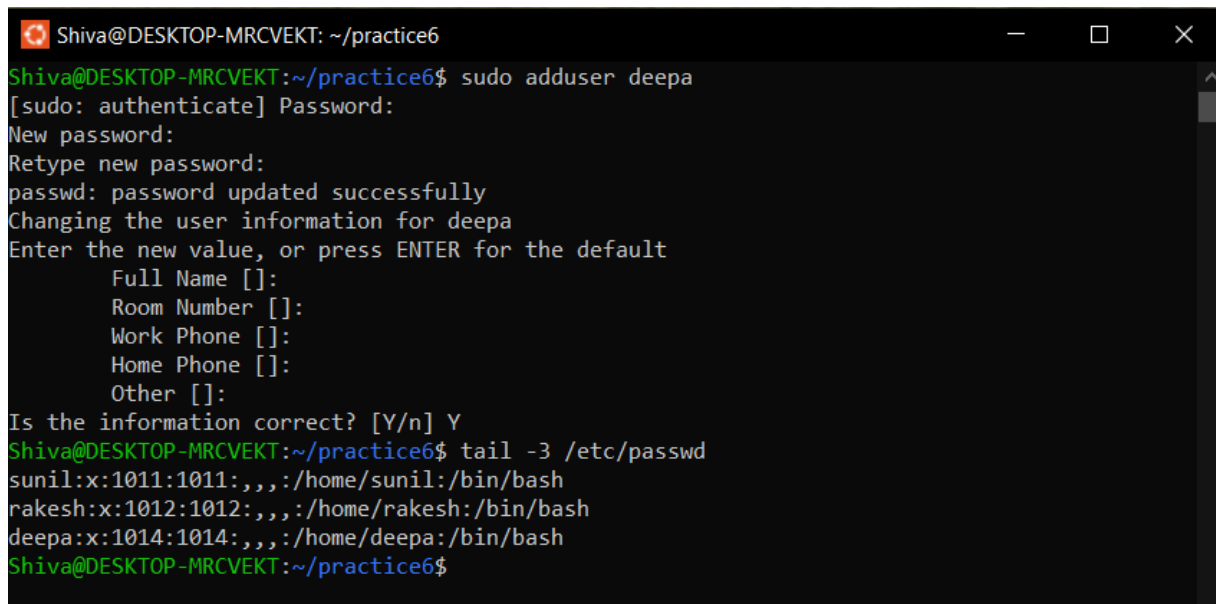
Administrative commands

From the notes, administrative (configuration) commands are used to create and manage users and groups. They change the system, so they need sudo in front. The first time I used sudo it asked for my own password. When I type a password in Linux nothing shows on the screen, not even stars, which surprised me the first time.

Creating a user

```
sudo adduser deepa
```

```
tail -3 /etc/passwd
```



```
Shiva@DESKTOP-MRCVEKT: ~/practice6
Shiva@DESKTOP-MRCVEKT:~/practice6$ sudo adduser deepa
[sudo: authenticate] Password:
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for deepa
Enter the new value, or press ENTER for the default
  Full Name []:
  Room Number []:
  Work Phone []:
  Home Phone []:
  Other []:
Is the information correct? [Y/n] Y
Shiva@DESKTOP-MRCVEKT:~/practice6$ tail -3 /etc/passwd
sunil:x:1011:1011:,,,:/home/sunil:/bin/bash
rakesh:x:1012:1012:,,,:/home/rakesh:/bin/bash
deepa:x:1014:1014:,,,:/home/deepa:/bin/bash
Shiva@DESKTOP-MRCVEKT:~/practice6$
```

Screenshot 1: making the user deepa and checking /etc/passwd.

First sudo asked for my password. Then adduser asked for a new password for deepa, two times. After that it asked for Full Name, Room Number and some more details. These are optional, so I just pressed Enter for all of them, and Y at the end.

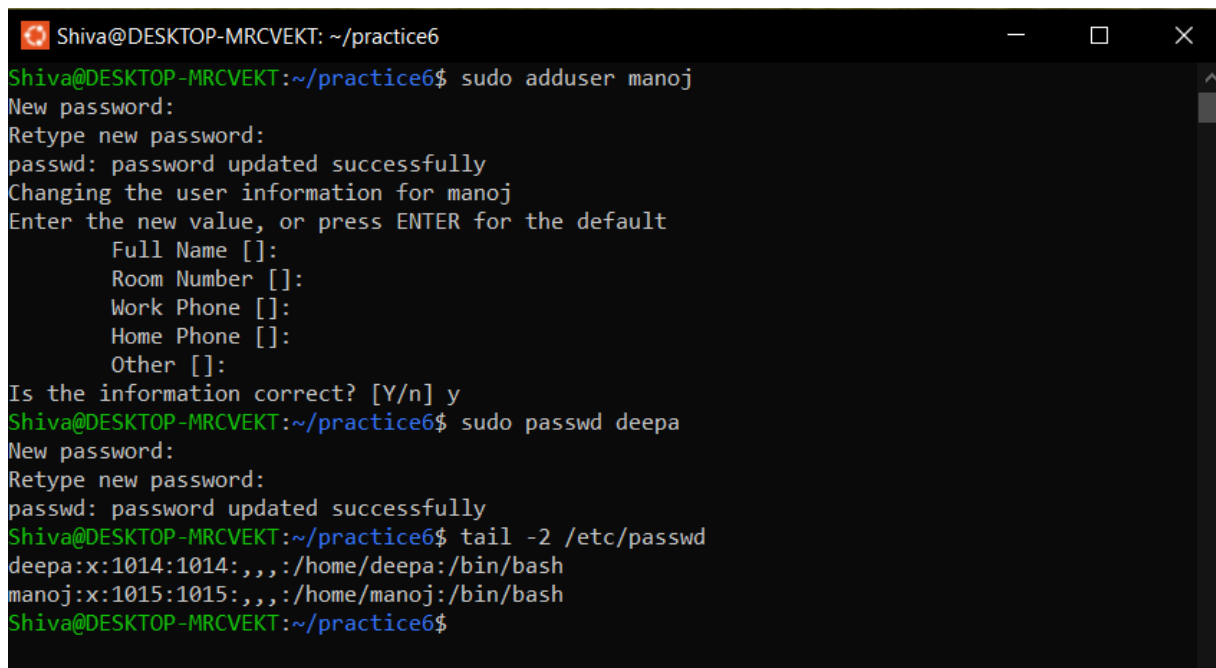
The notes say to check with cat /etc/passwd. That file has a line for every user on the system, including many system users, so it is very long. New users are added at the bottom, so I used tail -3 to see only the last 3 lines. deepa is there:

```
deepa:x:1014:1014:,,,:/home/deepa:/bin/bash
```

- deepa is the user name, and x means the password is kept in another safe file, not here.
- 1014 is the user id and the next 1014 is the group id. adduser also made a group called deepa for her.
- The empty commas are the Full Name and other details I skipped, then her home folder /home/deepa, and her shell /bin/bash.

A second user and changing a password

```
sudo adduser manoj  
sudo passwd deepa  
tail -2 /etc/passwd
```



```
Shiva@DESKTOP-MRCVEKT: ~/practice6  
Shiva@DESKTOP-MRCVEKT:~/practice6$ sudo adduser manoj  
New password:  
Retype new password:  
passwd: password updated successfully  
Changing the user information for manoj  
Enter the new value, or press ENTER for the default  
  Full Name []:  
  Room Number []:  
  Work Phone []:  
  Home Phone []:  
  Other []:  
Is the information correct? [Y/n] y  
Shiva@DESKTOP-MRCVEKT:~/practice6$ sudo passwd deepa  
New password:  
Retype new password:  
passwd: password updated successfully  
Shiva@DESKTOP-MRCVEKT:~/practice6$ tail -2 /etc/passwd  
deepa:x:1014:1014:,,,:/home/deepa:/bin/bash  
manoj:x:1015:1015:,,,:/home/manoj:/bin/bash  
Shiva@DESKTOP-MRCVEKT:~/practice6$
```

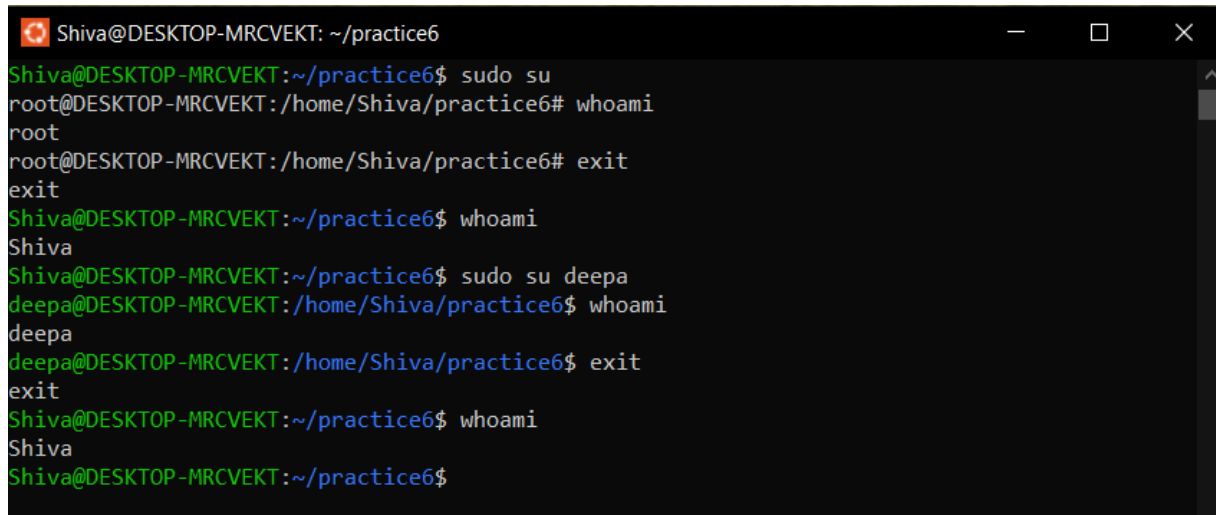
Screenshot 2: making manoj, changing deepa's password, and seeing both users.

This time sudo did not ask for my password again. It remembers it for a few minutes after the first time.

sudo passwd deepa is how to change a user's password. It asked for the new password two times and said password updated successfully. tail -2 shows both deepa (1014) and manoj (1015) are now in /etc/passwd.

Switching users

```
sudo su
whoami
exit
sudo su deepa
whoami
exit
```

A screenshot of a terminal window titled 'Shiva@DESKTOP-MRCVEKT: ~/practice6'. The terminal shows a sequence of commands and their outputs: 'sudo su' changes the prompt to 'root@DESKTOP-MRCVEKT:/home/Shiva/practice6#'; 'whoami' returns 'root'; 'exit' returns to the 'Shiva@DESKTOP-MRCVEKT:~/practice6\$' prompt. Then, 'sudo su deepa' changes the prompt to 'deepa@DESKTOP-MRCVEKT:/home/Shiva/practice6\$'; 'whoami' returns 'deepa'; 'exit' returns to the 'Shiva@DESKTOP-MRCVEKT:~/practice6\$' prompt. Finally, 'whoami' is run again, returning 'Shiva'. The terminal has a dark background with green and white text. The window has standard Linux window controls (minimize, maximize, close) in the top right corner.

```
Shiva@DESKTOP-MRCVEKT: ~/practice6
Shiva@DESKTOP-MRCVEKT:~/practice6$ sudo su
root@DESKTOP-MRCVEKT:/home/Shiva/practice6# whoami
root
root@DESKTOP-MRCVEKT:/home/Shiva/practice6# exit
exit
Shiva@DESKTOP-MRCVEKT:~/practice6$ whoami
Shiva
Shiva@DESKTOP-MRCVEKT:~/practice6$ sudo su deepa
deepa@DESKTOP-MRCVEKT:/home/Shiva/practice6$ whoami
deepa
deepa@DESKTOP-MRCVEKT:/home/Shiva/practice6$ exit
exit
Shiva@DESKTOP-MRCVEKT:~/practice6$ whoami
Shiva
Shiva@DESKTOP-MRCVEKT:~/practice6$
```

Screenshot 3: going into root and into deepa, and coming back with exit.

After `sudo su` the prompt changed to `root@DESKTOP-MRCVEKT` and the `$` at the end changed to `#`. The `#` means I am the root user, who can do anything on the system. `whoami` said `root`. `exit` brought me back and `whoami` said `Shiva` again.

`sudo su deepa` made me `deepa`. The prompt started with `deepa@` and `whoami` said `deepa`. `exit` brought me back to `Shiva`. So `exit` is how to come out, both from `root` and from another user.

Creating a group and adding users to it

```

sudo addgroup book-club
tail -2 /etc/group

sudo usermod -G book-club deepa
sudo usermod -G book-club manoj

grep book-club /etc/group

groups deepa

```

```

Shiva@DESKTOP-MRCVEKT: ~/practice6
Shiva@DESKTOP-MRCVEKT:~/practice6$ sudo addgroup book-club
Shiva@DESKTOP-MRCVEKT:~/practice6$ tail -2 /etc/group
manoj:x:1015:
book-club:x:1016:
Shiva@DESKTOP-MRCVEKT:~/practice6$ sudo usermod -G book-club deepa
Shiva@DESKTOP-MRCVEKT:~/practice6$ sudo usermod -G book-club manoj
Shiva@DESKTOP-MRCVEKT:~/practice6$ grep book-club /etc/group
book-club:x:1016:deepa,manoj
Shiva@DESKTOP-MRCVEKT:~/practice6$ groups deepa
deepa : deepa book-club
Shiva@DESKTOP-MRCVEKT:~/practice6$

```

Screenshot 4: making book-club and putting deepa and manoj in it.

After addgroup, the last line of /etc/group was book-club:x:1016: with nothing after the last colon, because the group had no members yet. The line above it is manoj, the group adduser made for him.

Then I used usermod -G for each user. Now the line is book-club:x:1016:deepa,manoj, so the members are written at the end, separated by a comma. I used grep to see only that line instead of the whole file. groups deepa also shows she is in deepa and book-club.

One thing to be careful about: usermod -G sets the full list of extra groups for the user. If deepa was already in some other group, -G book-club would take her out of it. To add a group and keep the old ones, the command is usermod -aG, where a means append.

Permissions - how to read ls -l

Every file, folder and link has permissions for three kinds of people. From the notes:

- User: the owner of the file.
- Group: the members of the file's group.
- Others: everyone else, who is not the owner and not in the group.

In ls -l the first letter tells the type: d for a folder, - for a file and l for a link. After that come 3 letters for the user, 3 for the group and 3 for others. r is read, w is write and x is execute. A dash means that permission is not there.

Each permission also has a number: r = 4, w = 2, x = 1. Adding them gives one number from 0 to 7 for each of user, group and others:

4 (r)	2 (w)	1 (x)	Number	Means
0	0	0	0	nothing
0	0	1	1	execute only
0	1	0	2	write only
0	1	1	3	write and execute

4 (r)	2 (w)	1 (x)	Number	Means
1	0	0	4	read only
1	0	1	5	read and execute
1	1	0	6	read and write
1	1	1	7	read, write and execute

Changing permissions with numbers

I made two empty files and tried the numbers from the table in the notes on file1.txt, checking with ls -l every time.

```
touch file1.txt file2.txt
chmod 644 file1.txt
ls -l
chmod 777 file1.txt
chmod 700 file1.txt
chmod 440 file1.txt
chmod 414 file1.txt
```

```
Shiva@DESKTOP-MRCVEKT: ~/practice6
Shiva@DESKTOP-MRCVEKT:~/practice6$ chmod 644 file1.txt
Shiva@DESKTOP-MRCVEKT:~/practice6$ ls -l
total 0
-rw-r--r-- 1 Shiva Shiva 0 Oct  8 17:44 file1.txt
-rw-r--r-- 1 Shiva Shiva 0 Oct  8 17:44 file2.txt
Shiva@DESKTOP-MRCVEKT:~/practice6$ chmod 777 file1.txt
Shiva@DESKTOP-MRCVEKT:~/practice6$ ls -l file1.txt
-rwxrwxrwx 1 Shiva Shiva 0 Oct  8 17:44 file1.txt
Shiva@DESKTOP-MRCVEKT:~/practice6$ chmod 700 file1.txt
Shiva@DESKTOP-MRCVEKT:~/practice6$ ls -l file1.txt
-rwx----- 1 Shiva Shiva 0 Oct  8 17:44 file1.txt
Shiva@DESKTOP-MRCVEKT:~/practice6$ chmod 440 file1.txt
Shiva@DESKTOP-MRCVEKT:~/practice6$ ls -l file1.txt
-r--r----- 1 Shiva Shiva 0 Oct  8 17:44 file1.txt
Shiva@DESKTOP-MRCVEKT:~/practice6$ chmod 414 file1.txt
Shiva@DESKTOP-MRCVEKT:~/practice6$ ls -l file1.txt
-r----xr-- 1 Shiva Shiva 0 Oct  8 17:44 file1.txt
Shiva@DESKTOP-MRCVEKT:~/practice6$
```

Screenshot 5: 644, 777, 700, 440 and 414 on the same file.

New files start as -rw-r--r--, which is 644: I can read and write, the group and others can only read.

- 777 gave -rwxrwxrwx, everything for everybody. The name turned green, because green means it can be executed.
- 700 gave -rwx-----, everything for me and nothing for anyone else.
- 440 gave -r--r-----, read only for me and the group, nothing for others.
- 414 gave -r----xr--. This one looks strange: the user can read, the group can only execute, and others can read.

Here is the same as a table, like the one in the notes:

Number	User	Group	Others	ls -l shows
777	rwX	rwX	rwX	-rwxrwxrwx
700	rwX	---	---	-rwx-----
440	r--	r--	---	-r--r-----
414	r--	--x	r--	-r-----xr--
460	r--	rw-	---	-r--rw----

The notes use sudo chmod. I did not need sudo here, because I am the owner of these files, and the owner can always change the permissions of his own files.

The 460 question and the letter way

The notes ask how to give only read to the user, read and write to the group and nothing to others. That is 4 for the user, 4 + 2 = 6 for the group and 0 for others, so 460.

```
chmod 460 file2.txt
```

Then the letter way. u is user, g is group and o is others. + gives a permission and - takes it away. In the notes it is written 0+x, but it is the small letter o for others, not a zero. I started file1.txt from 444 so the changes are easy to see.

```
chmod 444 file1.txt
```

```
chmod o+x file1.txt
```

```
chmod u+w file1.txt
```

```
chmod u+x file1.txt
```

```
chmod u-x file1.txt
```

```

Shiva@DESKTOP-MRCVEKT: ~/practice6
Shiva@DESKTOP-MRCVEKT:~/practice6$ chmod 460 file2.txt
Shiva@DESKTOP-MRCVEKT:~/practice6$ ls -l file2.txt
-r--rw---- 1 Shiva Shiva 0 Oct  8 17:44 file2.txt
Shiva@DESKTOP-MRCVEKT:~/practice6$ chmod 444 file1.txt
Shiva@DESKTOP-MRCVEKT:~/practice6$ ls -l file1.txt
-r--r--r-- 1 Shiva Shiva 0 Oct  8 17:44 file1.txt
Shiva@DESKTOP-MRCVEKT:~/practice6$ chmod o+x file1.txt
Shiva@DESKTOP-MRCVEKT:~/practice6$ ls -l file1.txt
-r--r--r-x 1 Shiva Shiva 0 Oct  8 17:44 file1.txt
Shiva@DESKTOP-MRCVEKT:~/practice6$ chmod u+w file1.txt
Shiva@DESKTOP-MRCVEKT:~/practice6$ ls -l file1.txt
-rw-r--r-x 1 Shiva Shiva 0 Oct  8 17:44 file1.txt
Shiva@DESKTOP-MRCVEKT:~/practice6$ chmod u+x file1.txt
Shiva@DESKTOP-MRCVEKT:~/practice6$ ls -l file1.txt
-rwxr--r-x 1 Shiva Shiva 0 Oct  8 17:44 file1.txt
Shiva@DESKTOP-MRCVEKT:~/practice6$ chmod u-x file1.txt
Shiva@DESKTOP-MRCVEKT:~/practice6$ ls -l file1.txt
-rw-r--r-x 1 Shiva Shiva 0 Oct  8 17:44 file1.txt
Shiva@DESKTOP-MRCVEKT:~/practice6$

```

Screenshot 6: 460 on file2.txt, then o+x, u+w, u+x and u-x one by one on file1.txt.

file2.txt became -r--rw----, which is exactly what the question asked for.

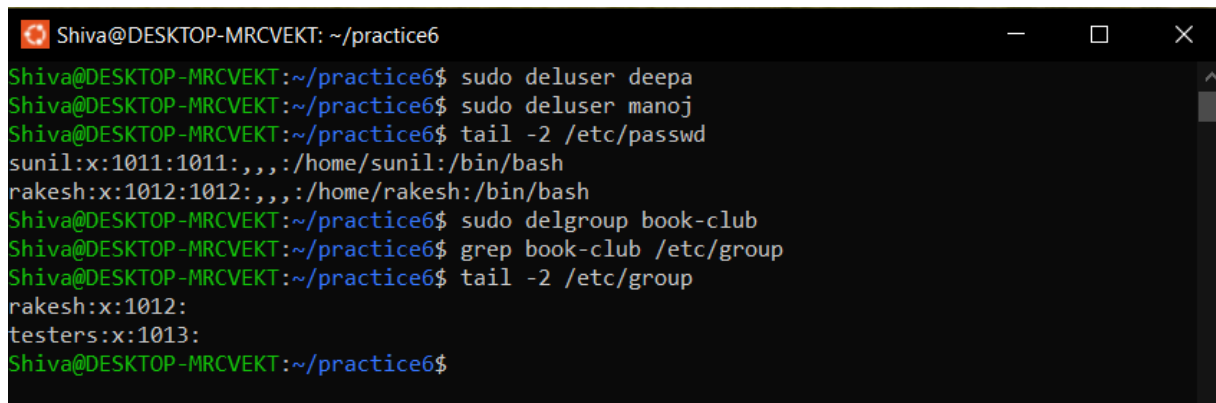
Command	What happened to file1.txt
chmod 444 file1.txt	Start: -r--r--r--, everyone can only read
chmod o+x file1.txt	-r--r--r-x, others got execute

Command	What happened to file1.txt
chmod u+w file1.txt	-rw-r--r-x, the user (me) got write
chmod u+x file1.txt	-rwxr--r-x, the user got execute
chmod u-x file1.txt	-rw-r--r-x, execute taken away from the user again

The good thing about the letter way is that it changes only one thing and leaves the rest as it was. With numbers I have to work out all three parts every time.

Deleting the users and the group

```
sudo deluser deepa
sudo deluser manoj
tail -2 /etc/passwd
sudo delgroup book-club
grep book-club /etc/group
tail -2 /etc/group
```



```
Shiva@DESKTOP-MRCVEKT: ~/practice6
Shiva@DESKTOP-MRCVEKT:~/practice6$ sudo deluser deepa
Shiva@DESKTOP-MRCVEKT:~/practice6$ sudo deluser manoj
Shiva@DESKTOP-MRCVEKT:~/practice6$ tail -2 /etc/passwd
sunil:x:1011:1011:,,,:/home/sunil:/bin/bash
rakesh:x:1012:1012:,,,:/home/rakesh:/bin/bash
Shiva@DESKTOP-MRCVEKT:~/practice6$ sudo delgroup book-club
Shiva@DESKTOP-MRCVEKT:~/practice6$ grep book-club /etc/group
Shiva@DESKTOP-MRCVEKT:~/practice6$ tail -2 /etc/group
rakesh:x:1012:
testers:x:1013:
Shiva@DESKTOP-MRCVEKT:~/practice6$
```

Screenshot 7: deepa, manoj and book-club are all gone.

deluser and delgroup did not print anything. To check, I looked at the files again. The last lines of /etc/passwd are now sunil and rakesh, the users that were there before, so deepa and manoj are gone. grep book-club printed nothing, so the group is gone too. Their own groups deepa and manoj also went away with them.

deluser removes the user but keeps the user's home folder by default. To remove the home folder also, there is `sudo deluser --remove-home <user-name>`.